



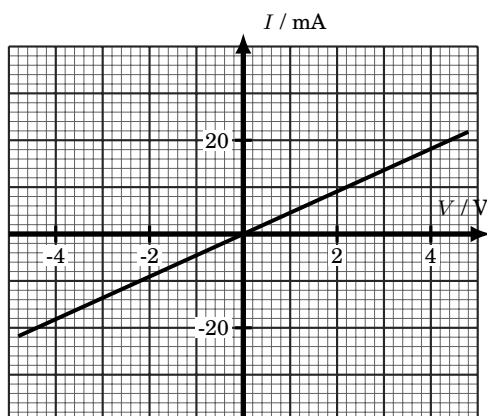
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 Date: _____

Mr. Rawson • GCSE Physics

Ohmic and Non-Ohmic Conductors

Three components, three shapes. Read the shape and you know the component.

1 · Ohmic conductors resistance stays the same



Graph A — a fixed resistor

The shape: a straight line through the origin.

Double the pd, and the current doubles too. So $R = \frac{V}{I}$ comes out **the same everywhere on the line**.

$$I \propto V$$

Examples

- a **fixed resistor**
- a **metal wire** — as long as it stays cool



The condition: at a constant temperature. Let it get hot and it stops being ohmic.

Worked example — the method

A point read off a *different* resistor's graph gives $V = 3.0 \text{ V}$ and $I = 25 \text{ mA}$.

Step 1 — convert the current to amps. $25 \text{ mA} = 25 \div 1000 = 0.025 \text{ A}$

Step 2 — divide. $R = \frac{V}{I} = \frac{3.0}{0.025} = 120 \Omega$

Forgetting step 1 gives 0.12 — a thousand times too small. The current axis on this sheet is always in **milliamps**.

Finding resistance from any graph

Pick a point on the line. Read V across. Read I up. Divide. $R = \frac{V}{I}$ — this works on every graph on this sheet.

The check: do it again at a second point. Same answer → the resistance is **constant** (ohmic). Different answer → the resistance has **changed**.

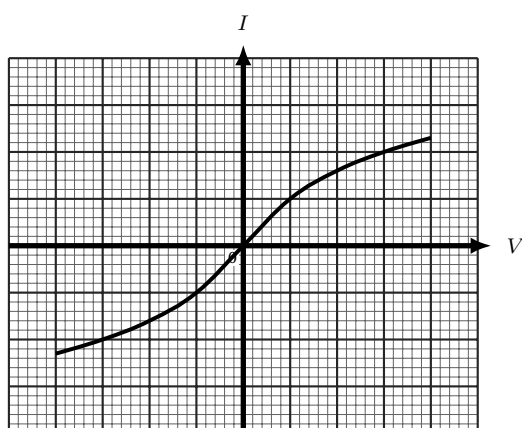
Exam-style question [4 marks]

Graph A is a real fixed resistor. Use the graph to find its resistance.

Choose a point where the line crosses a grid line *exactly* — it makes the reading easier.

The point I read: $V = \underline{\hspace{2cm}} \text{ V}$ $I = \underline{\hspace{2cm}} \text{ mA}$

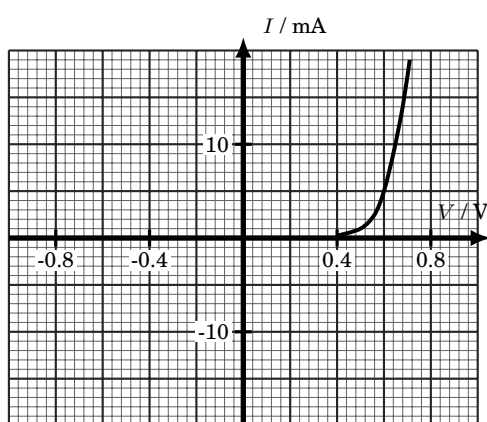
The resistance of a **lamp**, **diode**, **thermistor** or **LDR** is **not** constant — so their graphs are not straight lines.



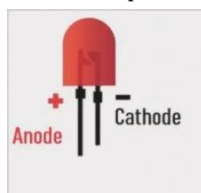
Graph B



the filament and its support wires

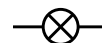


Graph C — a diode
note the scale: 1 square = 0.2 V



an LED is a diode — it only works one way round

2a The filament lamp



The shape: bends over towards the V axis — the same way on *both* sides.

Why: bigger current $\rightarrow$ filament **hotter** $\rightarrow$ ions **vibrate more** $\rightarrow$ more **collisions** $\rightarrow$ **resistance increases**.

Examples: a torch bulb, a car headlamp, an old-style light bulb.

Exam-style question [3 marks]

Explain why the resistance of a filament lamp increases as the potential difference across it increases.

2b The diode



The shape: flat along the axis one way, then rising steeply the other way.

Why: a diode lets current through in **one direction only**. Forwards, it conducts once the pd passes about 0.6 V. Backwards, its resistance is **very high**, so the current is almost zero.

Examples: an **LED** is a diode — connect it the wrong way round and it simply will not light. Also the charging circuit in a phone, a solar panel, and anything that must not be connected backwards.

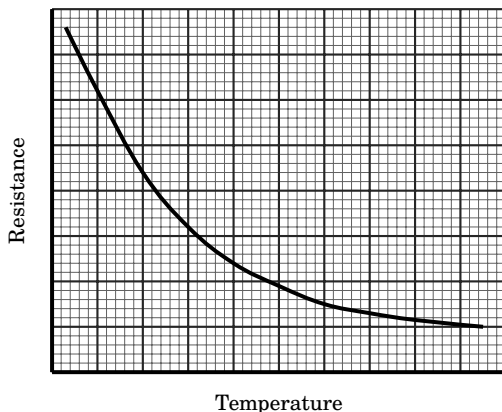
Exam-style question [2 marks]

A diode is connected in a circuit with a cell and a lamp, and the lamp lights. The diode is then turned round. Describe and explain what happens to the lamp.

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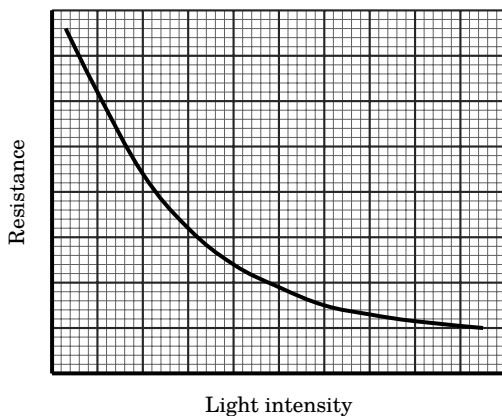
Both components below are non-ohmic too — but their graphs use **different axes**. Instead of I against V , we plot **resistance** against the thing the component responds to.



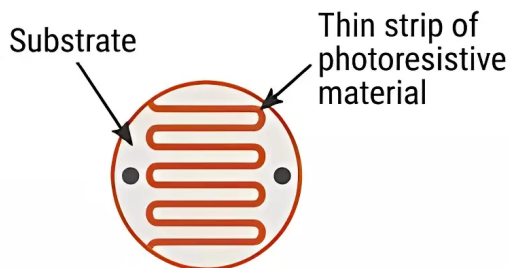
Graph D



the black bead at the top is the temperature sensor



Graph E



Light Dependent Resistor

the track is the part that responds to light

2c The thermistor



The shape: resistance **falls** as the temperature **rises** — steeply at first, then levelling off.

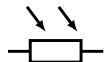
Remember it as: *hotter* → *lower resistance* → *bigger current*.

Examples: a thermostat, a fire alarm, the temperature sensor in an oven or a car engine.

Exam-style question [3 marks]

A thermistor is connected in series with a lamp and a cell. Describe and explain what happens to the brightness of the lamp as the room gets warmer.

2d The LDR



LDR stands for **light dependent resistor**.

The shape: the *same* falling curve as the thermistor — but against **light intensity**. Resistance **falls** as it gets **brighter**.

Remember it as: *brighter* → *lower resistance* → *bigger current*.

Examples: street lights that come on at dusk, automatic camera exposure, a phone screen that dims in a dark room.

Exam-style question [3 marks]

Explain how an LDR can be used to switch a street light on automatically when it gets dark.

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3 · Questions graphs A, B and C are on the front

1. State what is meant by an **ohmic conductor**. [2]

2. Name the component that produced each graph. [3]

A:

B:

C:

3. Use **graph B** to find the resistance of the lamp at $V = 1.0\text{ V}$ and at $V = 3.0\text{ V}$. Show your working, and say what has happened to the resistance. [4]

4. Jamie found the resistance for **graph A** by measuring its gradient, and wants to use the same method on **graph B**. State what Jamie should do instead for graph B, and explain why the method that worked on graph A does not carry across. [3]

5. Explain, in terms of the filament, **why** graph B bends. [3]

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6. Look at the **left-hand half** of the graphs, where the pd has been reversed. [4]

(a) Describe what **graph C** does, and explain why.

(b) **Graph B** is symmetrical about the origin. What does that tell you about a filament lamp?

7. A fixed resistor is left switched on until it becomes noticeably warm, and the test is repeated. Circle what happens to its graph, and explain in one sentence. [2]

it begins to bend over it stays perfectly straight it flattens along the axis

8. Sketch the graph you would expect for a **filament lamp**. Use both halves of the axes. [3]

